Exam Date & Time: 06-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER MAKEUP EXAMINATIONS, JAN 2025

MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-III [MAT 3135]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Missing data may be suitably assumed

1) Recall that the Mobius function μ is defined as

A) $$\mu(p_1{}^{r_1}p_2{}^{r_2}\cdots p_k{}^{r_k}) = \begin{cases}(-1)^k \text{ if } r_1 = r_2 = \cdots = r_k = 1 \\ 0 \qquad\qquad \text{otherwise}\end{cases},$$ (3)

where p_i are distinct primes and r_i are natural numbers. Let $f(n) = \sum_{d|n}\mu(d)$. What is the value of $f(144)$? Justify your answer.

B) Given any m integers none of which is a multiple of m , prove that two can be selected whose difference is a multiple of m (Here, $m \geq 2$). (3)

C) Let $d = (826, 1890)$. Use the Euclidean Algorithm to compute d, and then express d as a linear combination of 826 and 1890. (4)

2) Determine all the integers, if any, which simultaneously satisfy the following system of linear congruences: -

A) $$5x \equiv 1 \ (mod\ 6)$$ $$4x \equiv 13 \ (mod\ 15)$$ (3)

B) Determine all the odd primes p that could be represented by the integral binary quadratic form $x^2 + 2y^2$. Justify your answers. (3)

C) (i) Show that $7 \mid 3^{2n+1} + 2^{n+2}$ for all $n \in \mathbb{N}$. (4)

(ii) What are the last two digits in the ordinary decimal representation of 3^{400} ?

3) Determine whether the quadratic congruence $x^2 \equiv 150 \pmod{1009}$ is solvable or not. (3)

(Fact: 1009 is a prime number)

A)

B) Alice wants to secretly send the message "GO" using public key cryptographic techniques to Bob. Using the cipher below,

$$A \to 01, \quad B \to 02, \quad C \to 03, \quad \cdots, \quad Z \to 26$$

she converts her message into a natural number a and encrypts it using the public key $(k, m) = (17, 2773)$. Note that $2773 = 47 * 59$. Let b denote the number obtained after her encryption. (3)

(i) What is the key that Bob should use to decrypt it? In other words, find a positive integer h such that $b^h \equiv a \pmod{m}$.

(ii) Find the value of $b \pmod{2773}$.

(There is no need for ASCII table for solving this question)

C) Let G be a connected graph with at least 3 vertices. Then, show that G is bipartite if and only if it has no odd cycles. (4)

Give an example for a connected bipartite graph on 6 vertices with no cycles.

4) Prove that in a non-trivial self-complementary graph G, the number of vertices n is of the form 4k or $4k + 1, k \geq 1$. (3)

A) Draw a self-complementary graph of diameter 3.

B) Prove that in a tree on n vertices, the center of tree is either a single vertex or a pair of adjacent vertices. (3)

Draw a tree T for which both T and its complement $\overline{T}$ is connected.

C) Obtain the closeness centrality of all the vertices in (4)

(i) a cycle graph on n vertices

(ii) a complete bipartite graph, Kp,q

(iii) a path graph on 5 vertices.

5) Let G be a connected graph. Prove that G is Eulerian if and only if all the vertices are of even degree. (3)

A)

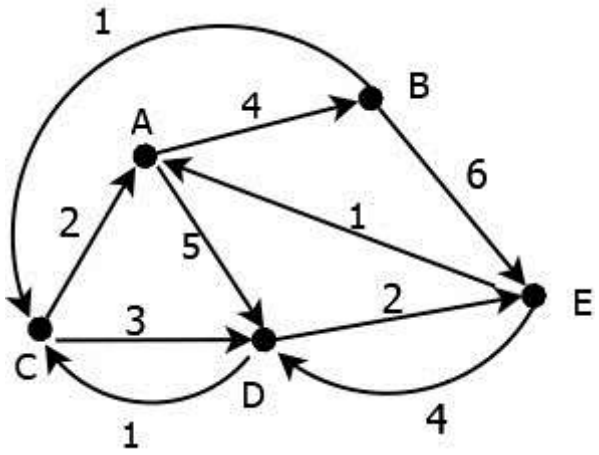
B) Obtain the determinant of adjacency matrix of (3)

(i) a cycle graph on 8 vertices

(ii) a path graph on 10 vertices
(iii) $H = K_4 \setminus e$, removal of an edge from K_4

by obtaining all the elementary spanning subgraphs and using the formula $\det(A(G)) = \sum(-1)^{n-c_1(H)-c(H)}\ 2^{c(H)}$ where summation runs over all elementary spanning subgraphs H of G and $c_1(H)$ and $c(H)$ are the number of components which are K_2's and cycles respectively.

C) Using Floyd Warshall algorithm, obtain all pairs shortest paths from every vertex to every other vertex for the graph as shown below. (4)



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